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Tiny TR-CAP: A novel small-scale benchmark dataset for general-purpose image captioning tasks

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ABSTRACT

In the last decade, the outstanding performance of deep learning has also led to a rapid and inevitable rise in automatic image captioning, as well as the need for large amounts of data. Although well-known, conventional and publicly available datasets have been proposed for the image captioning task, the lack of ground-truth caption data still remains a major challenge in the generation of accurate image captions. To address this issue, in this paper we introduced a novel image captioning benchmark dataset called Tiny TR-CAP, which consists of 1076 original images and 5380 handwritten captions (5 captions for each image with high diversity). The captions, which were translated into English using two web-based language translation APIs and a novel multilingual deep machine translation model, were tested against 11 state-of-the-art and prominent deep learning-based models, including CLIPCap, BLIP, BLIP2, FUSECAP, OFA, PromptCap, Kosmos2, MiniGPT4, LLaVA, BakLlaVA, and GIT. In the experimental studies, the accuracy statistics of the captions generated by the related models were reported in terms of the BLEU, METEOR, ROUGE-L, CIDEr, SPICE, and WMD captioning metrics, and their performance was evaluated comparatively. In the performance analysis, quite promising captioning performances were observed, and the best success rates were achieved with the OFA model with scores of 0.7097 BLEU-1, 0.5389 BLEU-2, 0.3940 BLEU-3, 0.2875 BLEU-4, 0.1797 METEOR, 0.4627 ROUGE-L, 0.2938 CIDEr, 0.0626 SPICE, and 0.4605 WMD. To support research studies in the field of image captioning, the image and caption sets of Tiny TR-CAP will also be publicly available on GitHub (https://github.com/abbasmemis/tiny_TR-CAP) for academic research purposes.

1. Introduction

In computer vision, automatic image captioning is basically defined as the task of describing the visual content of an image by computer systems and is also known as one of the staple, prominent, and challenging tasks in computer vision. Although this task stands out in the field of computer vision, it is actually at the intersection of image processing and natural language processing (NLP), since the visual content of images is expressed by text. Image describing and captioning are not, as is commonly known, the freshly minted tasks in computer science. In early studies focusing on image captioning, image descriptor systems using the template-based and retrieval-based approaches came to the fore [1]. The models built on template-based image captioning typically generate image captions by using predefined caption templates with a series of empty fields that are subsequently filled based on the visual content features of the image to be described, such as objects, object attributes, object relationships, and actions. In such approaches,

the accuracy of the caption produced is directly dependent on the accurate detection of objects, inter-object relations, and actions in the image. Moreover, the diversity of the captions produced is related to the predefined caption templates. On the other hand, retrieval-based approaches search for images in a training image pool that are most similar in content to the query image for which the output caption is to be generated and generate a caption from the descriptions of the retrieved candidate images. Although such methods can be able to generate syntactically correct captions, their ability to generate image-specific and semantically correct captions is quite limited. Furthermore, they often fail to generate more descriptive and diverse captions for the new query images due to the limited capacity of the training repository [2]. Even though the template-based and retrieval-based systems achieve certain success rates, they have significant drawbacks in automatic image captioning, and no satisfactory performance could be achieved in such systems [3].

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Over the past decade, we have witnessed the rise and unprecedented success of deep learning in computer vision. This breakthrough has also led to significant advances in the field of image captioning, as in many other research areas. The current literature on deep learning-based image captioning discusses various approaches such as convolutional neural networks (CNNs) [4], attention mechanisms [5], vision transformers [6], visual encoders [7], text decoders [8] and unsupervised learning strategies [9]. Deep learning-based methods can handle the complexities and challenges of image captioning quite well [1] and promise to compensate for the shortcomings of traditional approaches. In recent years, research on deep learning-based image captioning has increased, and it is noteworthy that several deep learning-based image captioning methods, approaches, and models have been proposed in these research studies [10–19]. Although the outstanding performance of deep learning-based methods has also led to a rapid and inevitable rise in automatic image captioning, they require large amounts of data. Well-known, conventional, and publicly available datasets have been proposed for the image captioning task, such as MS COCO [20], Flickr8K [21] and Flickr30K [22], but the lack of ground-truth caption data still remains a major challenge in the generation of accurate image captions. In order to address this issue, this paper presents a novel image captioning benchmark dataset called Tiny TR-CAP. The main contributions of the paper are threefold:

- A novel image-caption dataset, Tiny TR-CAP, consisting of 1076 images and 5380 handwritten captions, has been proposed for general-purpose image captioning tasks.
- We evaluated the performance of 11 different state-of-the-art deep learning-based caption generator models, which have become prominent in recent years, on the proposed image-caption dataset.
- The image captions written in Turkish were translated into English using 3 different language translation models, and we analyzed the caption generation performance of the models based on the language translation APIs (Application Programming Interfaces).

The rest of the paper is organized as follows: In Section 2, the proposed dataset Tiny TR-CAP is discussed. The methodological details of the study are explained in Section 3. In Section 4, the results of the experimental studies are presented. Discussions are given in Section 5. Finally, the conclusions are stated in Section 6.

2. Tiny TR-CAP image and caption dataset

Our novel image and caption benchmark dataset, Tiny TR-CAP, consists of 1076 images collected from different environments. The images in Tiny TR-CAP were acquired with a Samsung Galaxy S4 (GT-I9500) smartphone with an Android operating system running on it. The original images in JPEG format were acquired in RGB color space, and then the original images were resized with an aspect ratio of 0.4. The dimensions of most images in the Tiny TR-CAP image set are 461×819 and 819×461 , while the average width and height of all images are 603 and 667, respectively. Fig. 1 illustrates some sample images from the image set of Tiny TR-CAP.

In the image collection phase of the constructed dataset, it is intended to collect images from different environments, including a wide variety of objects and object relations. Following the image acquisition process, ground-truth (reference) captions were written for the images. This resulted in a total of 5380 handwritten captions, 5 for each image in the image set. As is well known, the task of generating ground-truth image captions is a very challenging, labor-intensive, and time-consuming process. Moreover, in order for the captioning models to generate image captions that can be considered accurate, the ground-truth captions representing the images to be used to train the models should describe the images in a clear, accurate, correct, concise, and understandable manner. Furthermore, since there may be

different ground-truth descriptions to describe an image, the creation of ground-truth captions may vary from person to person, i.e., the ground-truth captions to be created for the image are directly dependent on the person who will describe the image. In this context, and taking into account the aforementioned problems in image captioning, we have followed the below roadmap for the generation of ground-truth captions for the images in our proposed benchmark database:

The reference captions for all images in the Tiny TR-CAP caption dataset were handwritten by one person. Although it was initially thought that writing captions by a single person might reduce the diversity of captions, several approaches were taken to compensate for the negative effects that this situation might have on the captioning process. For this purpose, instead of a single caption, 5 captions were created for each image. However, in the process of creating 5 captions to describe each image, the aim was to make these captions different from each other. This was intended to increase the variety of captions. The fact that the process of generating the reference captions was carried out by one person also made it easier to control the diversity of the captions. Otherwise, when more than one person is involved in the captioning process, it should be noted that quite similar captions may be created by different people, resulting in caption duplication. Furthermore, captions written by different people may need to go through further meta-checking mechanisms to eliminate possible high similarity of texts.

One of the major challenges in image captioning is that an image can be described in many different ways, and all of these descriptions can be considered correct. For this reason, a large number of ground-truth captions are usually generated for the images in the caption databases. For the benchmark caption database proposed in this study, a similar strategy was followed, and 5 ground-truth captions were created for each image. Another feature of the proposed dataset is that some of the image captions are quite simple and some are quite detailed. Although it is challenging to create detailed ground-truth captions for simple and plain images (images with few objects and actions) and simple ground-truth captions for detailed images (images with many objects and actions), such a captioning protocol was defined to increase the variety of captions and add a different characteristic to the dataset.

The captions are originally written in Turkish, as Turkish is the native language of the authors. The average word length of all captions is approximately 20 (± 10) words, and the total number of unique words is 3801. All experimental analyses conducted in the study were performed on captions translated into English using different language translation APIs and deep translator models. In Fig. 2 the frequency distribution of the top 50 words in the Tiny TR-CAP caption set is visualized.

3. Methodologies

3.1. Data pre-processing

As the image descriptions in the Tiny TR-CAP were originally written in Turkish, all ground-truth image captions were translated into English with a pre-processing step. Two web-based language translation APIs (Google Translate API and Yandex Translate API) and a deep machine translation model (No Language Left Behind machine translation model, NLLB [23]) were used for this purpose. Google Translate API and Yandex Translate API are well-known and widely used web-based language translation applications. NLLB, on the other hand, is a recently proposed large-scale language translation model based on deep learning, with support for many languages. This machine translation model, based on a transformer structure [24], supports the translation of about 210 languages. Transformer models are also known to significantly improve translation quality [25]. The NLLB model is also based on a transformer structure. Fig. 3 shows the ground-truth Turkish captions for a sample image in the Tiny TR-CAP dataset and the corresponding English captions translated with the associated 3 language translators.

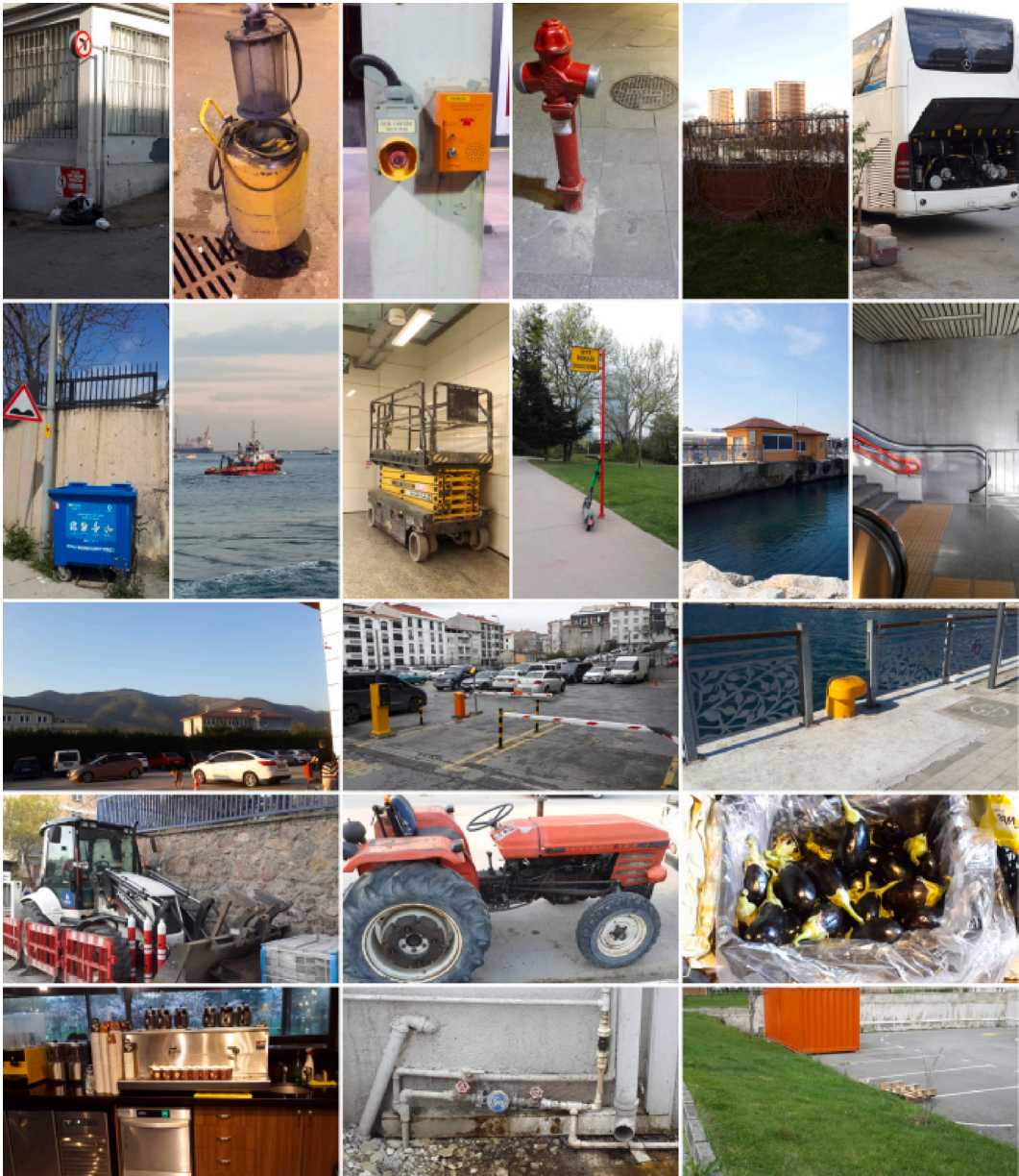


Fig. 1. Image samples from the Tiny TR-CAP image-caption dataset.

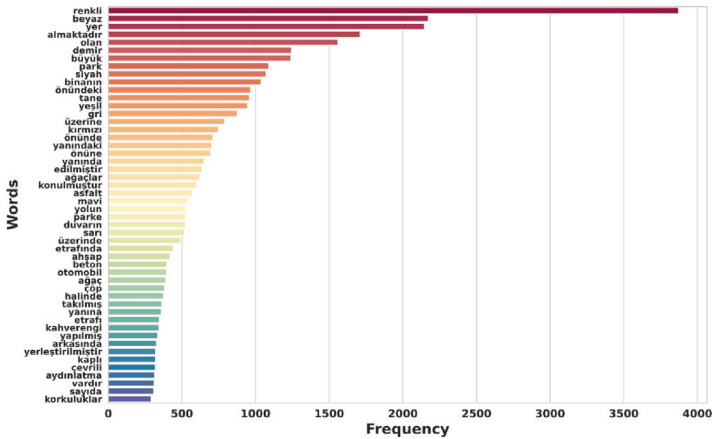


Fig. 2. Frequency distribution of the top-50 words in the Tiny TR-CAP caption set (Some common conjunctions and prepositions are excluded).



Original image captions in Turkish

- (1) Kapısı kapalı harabe bir akaryakıt istasyonu
- (2) Önüne tel örgülü demir bir kapı takılmış eski bir akaryakıt istasyonunda iki tane kamyon yer almaktadır
- (3) Önüne mavi tel örgülü demir bir kapı takılmış olan eski bir akaryakıt istasyonunda park halinde iki tane beyaz kamyon yer almaktadır
- (4) İçerisinde park halinde iki tane beyaz kamyon olan harabe durumdaki bir akaryakıt istasyonunun önüne mavi tel örgülü demir bir kapı takılmıştır ve zincirlenmiş demir kapının üzerinde bir park uyarı levhası yer alıyor
- (5) Harabe durumda olan bir akaryakıt istasyonunun yanında bir bina, ağaçlar, demir direkler ve park halinde beyaz renkli iki kamyon vardır ve akaryakıt istasyonunun önüne takılmış mavi tel örgülü demir kapıda bir zincir ve bir park uyarı levhası yer alıyor

Translated captions (Google Translate API)

- (1) A ruined gas station with closed doors
- (2) There are two trucks in an old gas station with a wire-mesh iron door in front of it
- (3) There are two white trucks parked at an old gas station with a blue wire mesh iron gate in front of it
- (4) An iron gate with blue wire mesh has been installed in front of a dilapidated gas station with two white trucks parked inside, and a parking warning sign is placed on the chained iron gate
- (5) There is a building, trees, iron poles and two parked white trucks next to a dilapidated fuel station, and there is a chain and a parking warning sign on the blue wire mesh iron gate installed in front of the fuel station

Translated captions (Yandex Translate API)

- (1) The door closed gas station ruins
- (2) Old iron gate in a fence in front with an attached gas station is located in two truck
- (3) The blue wire with the iron stuck in a door in front braided in two white truck parked in the old gas station is located
- (4) The white truck in the parking lot in front of the ruined state in a mobile station, the two braided iron door iron gate fitted to the blue wire and chained in a park on the warning signs may include
- (5) A mobile station which is in a poor state in the heart of a building, trees, poles and there are two trucks in the parking lot of iron in the color white, and blue wire braided fuel station stuck to the front of the warning signs of iron include a chain on the door and a park

Translated captions (No Language Left Behind (NLLB) deep machine translation model)

- (1) A gas station in ruins closed
- (2) Two trucks are standing in an old gas station with a wired iron door
- (3) Two white trucks are parked in an old gas station with a blue wire iron door
- (4) Inside a ruined gas station with two white trucks parked in a blue-wire iron gate and a parking warning sign on a chained iron gate
- (5) Next to a ruined gas station are two white trucks with a building, trees, iron posts, and a parking lot, and a chain and a parking warning sign on the front of the gas station with a blue wire-woven iron gate

Fig. 3. The ground-truth Turkish captions for a sample image in the Tiny TR-CAP and the corresponding English captions translated using the Google Translate API, the Yandex Translate API and the No Language Left Behind (NLLB) deep machine translation model.

3.2. Caption generation models

In the field of image captioning, research can be categorized into two main areas: (i) architectural innovations and (ii) dataset improvements. Architectural research aims to improve the image-text representation, whereas dataset-focused studies aim to construct cleaner, more detailed, and larger datasets. In this context, the study proposed in this paper can be included in the second category of dataset improvements. As the study focused on the construction of a novel image-caption dataset, we did not research a specific study on architectural innovations. However, we evaluated 11 recently proposed and prominent deep learning-based methods for image captioning, including CLIPCap [26], BLIP [27], BLIP2 [28], FUSECAP [29], OFA [30], PromptCap [31], Kosmos2 [32], MiniGPT4 [33], LLaVA [34], BakLLaVA [35] and GIT [36], on the proposed dataset.

The CLIPCap [26] introduces a lightweight, efficient approach to image captioning for both visual and textual processing. It uses the Contrastive Language-Image Pre-Training (CLIP) model's [37] embeddings as a prefix for text embedding and fine-tunes a language model to generate captions. The language model consists of GPT-2 architecture [38].

This method allows for faster training and lower model complexity without the need for large datasets or annotations.

The BLIP (Bootstrapping Language-Image Pre-training) [27] methodology is designed for unified vision-language understanding and generation. BLIP employs an encoder-decoder architecture. It integrates a Vision Transformer (ViT) [39] as an image encoder to process images. In parallel, its uni-modal text encoder mirrors the BERT [40] structure to generate feature vectors from input text. The image-grounded text encoder deviates from traditional BERT architectures by incorporating cross-attention within its transformer blocks, a change that replaces casual self-attention layers with bi-directional self-attention layers for improved context understanding. The architecture is further complemented by an image-grounded text decoder, which is designed as a text decoder featuring Causal Self-Attention, cross-attention mechanisms, and a feed-forward network. It proposes different architectural structures for different tasks, and this is the weakest aspect of the study.

FUSECAP [29] has been developed with the aim of generating significantly more detailed captions. In the absence of a properly customized dataset, it innovatively augments existing captions by incorporating visual details derived from object detectors, attribute recognizers, and Optical Character Recognizers (OCR). The additional information is then transformed into enriched captions using a specialized large language model (LLM) fuser. FlanT5-XL [41] has been adopted as the LLM fuser, specifically trained on a small dataset created using Chat GPT-3 [42]. It uses the BLIP as its base architecture. This technique improves captions by including additional information, resulting in more descriptive and detailed captions.

OFA [30], which stands for "One For All", is a sequence-to-sequence model designed to serve as an omni-model for multimodal tasks. It is the first attempt to combine vision and language, vision-only, and language-only tasks into a single framework. The architecture of OFA is based on the encoder-decoder concept. The transformer decoder consists of self-attention and feed-forward networks, and it utilizes cross-attention mechanisms to connect the decoder and encoder outputs.

PromptCap [31] leverages OFA as its base architecture, with significant contributions from the dataset training methodology. The approach claims that more detailed captions can be generated using visual question-answering datasets. To generate new captions, PromptCap employs GPT-3 to synthesize them and highlights the use of language models in image captioning tasks. The model starts its training process with an OFA checkpoint. Subsequent fine-tuning with the newly proposed dataset enables PromptCap to outperform the original OFA model. This method not only highlights the importance of innovative dataset utilization but also demonstrates the value of combining advanced linguistic models to improve the field of image captioning.

BLIP-2 [28] improves the efficiency of vision-language pre-training by bootstrapping from frozen image encoders and language models, using a Querying Transformer (Q-Former) for modality bridging. The architecture achieves efficiency through a design with fewer trainable parameters, leveraging a frozen image encoder and an LLM. The Q-Former architecture consists of two transformer sub-modules that share the same self-attention layers. One sub-module is dedicated to handling the output of the image encoder, while the other acts as the input text decoder. The Q-former is initialized using the pre-trained weights from the BERT base. BLIP 2 utilizes CLIP as the image encoder for the input images. For the decoder, it integrates models from the OPT family [43] for decoder-specific tasks and the FlanT5 family [44] for encoder-decoder tasks.

Kosmos-2 [32] is a multi-modal large language model that aims to connect text to the visual world and improve human-AI interaction for vision-language tasks. It introduces GRIT, a web-scale dataset of grounded image-text pairs, and employs it in training alongside multimodal corpora. Kosmos-2 uses the same model architecture and training standards as Kosmos-1. The vision encoder consists of 24

layers, and the multi-modal large language model component is a 24-layer MAGNETO Transformer [45]. The total number of trainable parameters is 1.6 billion.

The MiniGPT-4 [33] methodology improves vision-language understanding by aligning a pre-trained visual encoder with a large language model. The MiniGPT-4 architecture consists of ViT, Q-Former, a single linear projection layer, and an advanced Vicuna LLM. The image feature extraction parts, ViT and Q-Former, are derived from BLIP-2. MiniGPT-4 only needs to train the linear projection layer to align the visual features with the Vicuna.

The Generative Image-to-Text Transformer (GIT) methodology [36] simplifies vision-language tasks by using a single architecture with an image encoder and a text decoder. It focuses on efficiently mapping images to their textual descriptions without using complex structures. The model consists of a ViT image encoder and a BERT-like transformer decoder.

The Large Language and Vision Assistant (LLaVA) methodology [34] introduces a novel approach to multi-modal understanding by bridging the gap between LLMs and visual data. It consists of a huge Vicuna LLM as a text generator and a pre-trained CLIP ViT as a visual encoder. Image feature vectors are obtained by the visual encoder and converted into image tokens by the feed-forward network. The LLM takes image and text tokens and generates text. BakLLaVa [35] is just the LLaVa trained on more datasets.

3.3. Caption evaluation metrics

Since the predicted captions produced by caption generation models must not only match the reference (ground-truth) captions syntactically but also semantically, measuring the success of image captioning models is quite difficult, in contrast to other computer vision tasks. Therefore, several evaluation metrics already proposed in the current image annotation literature are generally used instead of a single standard metric representing the overall captioning performance [46,47]. In the performance evaluation of image captioning systems, metrics of BLEU (BiLingual Evaluation Understudy) [48], METEOR (Metric for Evaluation of Translation with Explicit Ordering) [49], ROUGE-L (Recall-Oriented Understudy for Gisting Evaluation) [50] and CIDEr (Consensus-based Image Description Evaluation) [51] stand out in general [52]. Besides, SPICE (Semantic Propositional Image Caption Evaluation) [53] and WMD (Word Mover's Distance) [54] are also among the commonly used caption evaluation metrics. Therefore, we employed the six related caption metrics, BLEU, METEOR, ROUGE-L, CIDEr, SPICE, and WMD, to measure the performance of the methods tested on our proposed benchmark dataset. The BLEU score, which is one of the most widely used quantification metrics, is calculated by taking the geometric mean of the precision scores produced on an n-gram basis for the predicted captions. The METEOR score, another n-gram-based caption quantification metric, examines the WordNet-based similarity and word root matching and considers the recall statistic. The metric of ROUGE-L, which is an LCS-based (longest common subsequence) caption quantification metric, basically measures the F1 score according to the longest matching word sequence between the ground truth and the predicted captions and gives the similarity rate between the two texts. Similar to the BLEU and METEOR metrics, the CIDEr metric is also based on n-grams. However, unlike the other metrics, it weights each n-gram using the TF-IDF (Term Frequency-Inverse Document Frequency) method, and therefore it increases the weight of n-grams that occur frequently in the reference caption but less frequently in the whole dataset and highlights this kind of word. As noted in [54], the WMD distance measures the dissimilarity between two captions as the minimum distance that the words in the generated caption need to travel to reach the words in the reference captions. Unlike metrics based on n-gram overlap, SPICE is an automated caption evaluation metric defined over scene graphs. It is known to be more robust and dominant in simulating human judgment, as it measures the quality of generated captions by analyzing their semantic content.

3.4. Environmental setup

For the experimental analysis of the proposed study, a computer system with an Intel i9-12900K CPU, an NVIDIA GeForce RTX 3090 Ti graphics card, 32 GB of memory, and running the Ubuntu-Linux operating system was employed. The software-based implementation used the PyTorch deep learning library (v2.0) [55] with the Python programming language (v3.8). Three different language translators were used to obtain the English captions: the Google Translate API, the Yandex Translate API, and the NLLB deep machine translation model. In addition to these technical and environmental details, the parametric values set for the related 11 deep captioning models are detailed in Table 1.

4. Results

In the experimental studies, caption predictions were generated for each image in the proposed dataset using 11 different deep learning-based methods, which are described in detail in the methodologies section. As mentioned in detail in Section 2, ground-truth captions were written in Turkish for the images in the Tiny TR-CAP dataset, and these ground-truth captions were translated into English using the Google Translate API, the Yandex Translate API, and the NLLB deep machine translation model. In the first stage of the experimental studies, we compared the captions predicted by the models with the ground-truth captions obtained using each of the aforementioned translation APIs and analyzed the performance of the models in caption generation based on the language translate API. The performance statistics of the 11 related deep captioning models for different language translators are presented in Table 2 as follows.

In Table 2, the performance rank numbers are written next to the three most successful models, and the highest value for each metric is highlighted in bold. Considering the caption generation success rates in Table 2, the most successful models for each language translator are OFA, BLIP, and PromptCap. In addition, the overwhelming majority of the highest metric values were achieved by the NLLB deep machine translation model. To analyze the individual performance of the models in the caption generation process, the mean performance values of each model in the three language translator models were calculated, and the observed mean performance statistics are given in Table 3 on a caption generator model basis.

Similar to Table 2, the three most successful models are numbered, and the highest mean for each metric is highlighted in bold in Table 3. Focusing on the mean success rates in Table 3, it can be seen that the most successful deep caption generator models are OFA, PromptCap, and BLIP, respectively. Additionally, the lowest performance was observed in CLIPCap and LLaVA caption models in general. In addition to the scores of the performance metrics in Tables 2 and 3, the image captions produced by all the models for 4 sample images selected from the Tiny TR-CAP dataset are presented in Fig. 4.

In the second phase of the experimental studies, as an ablation study, the ground-truth captions of all language translator models were concatenated, and the predicted captions generated by the models were tested on this merged caption set. The results observed in the related ablation study are presented in Table 4. As expected, the diversity of the combined caption set significantly improved model performance. In the set where the ground-truth captions of all language translator models were combined, it was observed that the most successful deep caption generator models are OFA, PromptCap, and BLIP.

5. Discussions

The current literature of automatic image captioning includes well-known, conventional, and publicly available datasets such as MS COCO and Flickr for general image captioning tasks. Apart from these well-known datasets, several other image-caption sets consisting of English

Table 1
Model parameters.

Model	Parameter set
CLIPCap [26]	Repo_id="akhaliq/CLIP-prefix-captioning-conceptual-weights", filename="conceptual_weights.pt"
BLIP [27]	Pretrained="model_large_caption.pth", image_size=384, vit='large', sample=False, num_beams=3, max_length=20, min_length=5
BLIP2 [28]	"Salesforce/blip2-opt-2.7b", load_in_8bit=True,
FUSECAP [29]	Default
OFA [30]	"Bpe_dir": "utils/BPE", "eval_cider": False, "beam": 5, "max_len_b": 16, "no_repeat_ngram_size": 3, "seed": 7
PromptCap [31]	"Tifa-benchmark/promptcap-coco-vqa", "what does the image describe?"
Kosmos2 [32]	<Grounding>An image of
MiniGPT4 [33]	Prerained_minigpt4_7b.pth, arch: mini_gpt4, model_type: pretrain_vicuna0, num_beams=3, temperature=1, max_new_tokens=40, max_length=1000, prompt : "Describe this image."
LlaVA [34]	Llava-hf/llava-1.5-7b-hf, USER: <image>\nWhat are these?\nASSISTANT:
BakLlaVA [35]	Llava-hf/bakllava-v1-hf, USER: <image>\nWhat are these?\nASSISTANT:
GIT [36]	Microsoft/git-large-coco , max_length=50

Table 2
Performance of the deep captioning models for different language translators.

Translator	Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	SPICE	WMD
Google Translate API	CLIPCap [26]	0.3071	0.1794	0.0953	0.0540	0.0658	0.2186	0.0812	0.0466	0.3436
	BLIP [27]	0.5954 ²	0.4057 ²	0.2782 ³	0.1969 ³	0.1497	0.4039 ³	0.3007 ²	0.1032	0.4352 ³
	BLIP2 [28]	0.5312	0.3591	0.2423	0.1678	0.1382	0.3768	0.2826	0.0978	0.4246
	FUSECAP [29]	0.5114	0.3156	0.1902	0.1186	0.1763 ²	0.3592	0.1490	0.1451¹	0.4230
	OFA [30]	0.6306 ¹	0.4353 ¹	0.2998 ¹	0.2094 ¹	0.1602 ³	0.4151 ¹	0.3099¹	0.1124 ³	0.4414¹
	PromptCap [31]	0.5745 ³	0.3998 ³	0.2812 ²	0.2012 ²	0.1513	0.4104 ²	0.2990 ³	0.1088	0.4402 ²
	Kosmos2 [32]	0.5412	0.3535	0.2189	0.1385	0.1343	0.3675	0.2542	0.0978	0.4179
	MiniGPT4 [33]	0.4769	0.2929	0.1789	0.1136	0.1881¹	0.3491	0.1249	0.1324 ²	0.4132
	LlaVA [34]	0.2920	0.1557	0.0861	0.0513	0.1473	0.2565	0.0407	0.1069	0.3761
	BakLlaVA [35]	0.4616	0.2611	0.1517	0.0915	0.1395	0.3039	0.1574	0.0978	0.4039
	GIT [36]	0.5506	0.3600	0.2344	0.1577	0.1358	0.3687	0.2556	0.0943	0.4174
Yandex Translate API	CLIPCap [26]	0.3028	0.1934	0.1103	0.0680	0.0704	0.2251	0.0639	0.0416	0.3416
	BLIP [27]	0.5775 ²	0.3996 ²	0.2776 ²	0.1972 ²	0.1409	0.3588 ³	0.2250 ³	0.0869	0.4254 ³
	BLIP2 [28]	0.5109	0.3481	0.2288	0.1551	0.1295	0.3372	0.2185	0.0794	0.4159
	FUSECAP [29]	0.5191	0.3101	0.1783	0.1078	0.1678 ²	0.3292	0.1201	0.1197 ¹	0.4172
	OFA [30]	0.6164 ¹	0.4322 ¹	0.2937 ¹	0.2016 ¹	0.1505 ³	0.3750 ¹	0.2481 ¹	0.0929	0.4322 ¹
	PromptCap [31]	0.5515 ³	0.3889 ³	0.2703 ³	0.1906 ³	0.1412	0.3627 ²	0.2330 ²	0.0891	0.4301 ²
	Kosmos2 [32]	0.5159	0.3394	0.2071	0.1289	0.1259	0.3290	0.1991	0.0801	0.4088
	MiniGPT4 [33]	0.4831	0.2917	0.1735	0.1078	0.1791 ¹	0.3343	0.1028	0.1129 ²	0.4069
	LlaVA [34]	0.2904	0.1549	0.0820	0.0469	0.1435	0.2334	0.0301	0.0930 ³	0.3706
	BakLlaVA [35]	0.4385	0.2469	0.1394	0.0828	0.1295	0.2646	0.1201	0.0816	0.3947
	GIT [36]	0.5306	0.3516	0.2224	0.1450	0.1281	0.3345	0.1937	0.0767	0.4089
NLLB Translate model	CLIPCap [26]	0.3244	0.1917	0.1041	0.0610	0.0701	0.2251	0.0862	0.0432	0.3467
	BLIP [27]	0.6125 ²	0.4175 ³	0.2866 ³	0.2031 ³	0.1522	0.4062 ³	0.3011 ³	0.0943	0.4327 ³
	BLIP2 [28]	0.5609	0.3846	0.2599	0.1818	0.1410	0.3850	0.2734	0.0872	0.4235
	FUSECAP [29]	0.5091	0.3138	0.1847	0.1130	0.1807 ²	0.3619	0.1307	0.1286 ¹	0.4198
	OFA [30]	0.6468¹	0.4463¹	0.3063¹	0.2151¹	0.1627 ³	0.4189¹	0.3067 ¹	0.1011 ³	0.4404 ¹
	PromptCap [31]	0.6054 ³	0.4211 ²	0.2958 ²	0.2136 ²	0.1560	0.4171 ²	0.3053 ²	0.0979	0.4399 ²
	Kosmos2 [32]	0.5494	0.3570	0.2200	0.1375	0.1344	0.3694	0.2402	0.0852	0.4140
	MiniGPT4 [33]	0.4569	0.2747	0.1639	0.1030	0.1854 ¹	0.3416	0.0917	0.1176 ²	0.4056
	LlaVA [34]	0.2745	0.1416	0.0741	0.0415	0.1414	0.2388	0.0287	0.0943	0.3688
	BakLlaVA [35]	0.4411	0.2434	0.1353	0.0778	0.1352	0.2877	0.1327	0.0843	0.3949
	GIT [36]	0.5723	0.3793	0.2504	0.1709	0.1385	0.3775	0.2555	0.0858	0.4171

caption texts exist. Table 5 compares some of the major English caption sets, along with statistics regarding the number of images and captions per image. Moreover, relatively limited caption sets written in some specific languages, such as Tiny TR-CAP, have also been published in recent years [56–60].

Although this study presents a completely novel dataset for general-purpose image captioning, and despite its contributions stated in the previous sections, it has some limitations and discussable aspects. The first issue that may be discussed as a limitation is the small size of the Tiny TR-CAP dataset compared to other caption datasets. The relatively small size of Tiny TR-CAP may seem to limit its applicability to larger image captioning tasks or real-world applications. Obviously, Tiny TR-CAP alone may not be sufficient for training large and comprehensive models and thus will not be sufficient for a custom captioning model to generate highly accurate, meaningful, and satisfactory captions. However, we expect that the Tiny TR-CAP dataset will contribute to building much more successful models when used in combination with such limited handwritten image-caption sets. In particular, it can contribute

to the fine-tuning of transferred and pre-trained models. Furthermore, since the proposed dataset is completely handwritten (not synthetic), it can be used directly for efficient training or fine-tuning of the models. The Tiny TR-CAP image and caption dataset is also proposed as a benchmark to test and validate the performance of the models.

Secondly, the number of captions per image in the dataset can also be discussed. In the Tiny TR-CAP dataset, 5 captions are provided for each image. Since the datasets constructed for image captioning tasks usually have an average of 5 captions, we have basically created 5 captions for each image in our proposed dataset. This is the main reason why we adopted such a strategy in deciding the number of captions defined per image. In addition, exceeding 5 captions generally leads to repetition of captions, and although there are minor syntactic differences in the captions, they do not lead to significant semantic variations. Besides, for each image, we first created short and concise captions and then increased the length and complexity of the initial captions. In this way, instead of a large number of similar captions, a set of 5 syntactically and semantically complex captions was obtained

Table 3
Model-based caption generation performances.

Model	Translator	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	SPICE	WMD
CLIPCap [26]	Google Translate API	0.3071	0.1794	0.0953	0.0540	0.0658	0.2186	0.0812	0.0466	0.3436
	Yandex Translate API	0.3028	0.1934	0.1103	0.0680	0.0704	0.2251	0.0639	0.0416	0.3416
	NLLB Translate Model	0.3244	0.1917	0.1041	0.0610	0.0701	0.2251	0.0862	0.0432	0.3467
	Mean (μ)	0.3114	0.1882	0.1032	0.0610	0.0688	0.2229	0.0771	0.0438	0.3440
BLIP [27]	Google Translate API	0.5954	0.4057	0.2782	0.1969	0.1497	0.4039	0.3007	0.1032	0.4352
	Yandex Translate API	0.5775	0.3996	0.2776	0.1972	0.1409	0.3588	0.2250	0.0869	0.4254
	NLLB Translate Model	0.6125	0.4175	0.2866	0.2031	0.1522	0.4062	0.3011	0.0943	0.4327
	Mean (μ)	0.5951 ²	0.4076 ²	0.2808 ³	0.1991 ³	0.1476	0.3896 ³	0.2756 ³	0.0948	0.4311 ³
BLIP2 [28]	Google Translate API	0.5312	0.3591	0.2423	0.1678	0.1382	0.3768	0.2826	0.0978	0.4246
	Yandex Translate API	0.5109	0.3481	0.2288	0.1551	0.1295	0.3372	0.2185	0.0794	0.4159
	NLLB Translate Model	0.5609	0.3846	0.2599	0.1818	0.1410	0.3850	0.2734	0.0872	0.4235
	Mean (μ)	0.5343	0.3639	0.2437	0.1682	0.1362	0.3663	0.2582	0.0881	0.4213
FUSECAP [29]	Google Translate API	0.5114	0.3156	0.1902	0.1186	0.1763	0.3592	0.1490	0.1451	0.4230
	Yandex Translate API	0.5191	0.3101	0.1783	0.1078	0.1678	0.3292	0.1201	0.1197	0.4172
	NLLB Translate Model	0.5091	0.3138	0.1847	0.1130	0.1807	0.3619	0.1307	0.1286	0.4198
	Mean (μ)	0.5132	0.3132	0.1844	0.1131	0.1749 ²	0.3501	0.1333	0.1311 ¹	0.4200
OFA [30]	Google Translate API	0.6306	0.4353	0.2998	0.2094	0.1602	0.4151	0.3099	0.1124	0.4414
	Yandex Translate API	0.6164	0.4322	0.2937	0.2016	0.1505	0.3750	0.2481	0.0929	0.4322
	NLLB Translate Model	0.6468	0.4463	0.3063	0.2151	0.1627	0.4189	0.3067	0.1011	0.4404
	Mean (μ)	0.6313 ¹	0.4379 ¹	0.3000 ¹	0.2087 ¹	0.1578 ³	0.4030 ¹	0.2882 ¹	0.1021 ³	0.4380 ¹
PromptCap [31]	Google Translate API	0.5745	0.3998	0.2812	0.2012	0.1513	0.4104	0.2990	0.1088	0.4402
	Yandex Translate API	0.5515	0.3889	0.2703	0.1906	0.1412	0.3627	0.2330	0.0891	0.4301
	NLLB Translate Model	0.6054	0.4211	0.2958	0.2136	0.1560	0.4171	0.3053	0.0979	0.4399
	Mean (μ)	0.5771 ³	0.4032 ³	0.2825 ²	0.2018 ²	0.1495	0.3967 ²	0.2791 ²	0.0986	0.4367 ²
Kosmos2 [32]	Google Translate API	0.5412	0.3535	0.2189	0.1385	0.1343	0.3675	0.2542	0.0978	0.4179
	Yandex Translate API	0.5159	0.3394	0.2071	0.1289	0.1259	0.3290	0.1991	0.0801	0.4088
	NLLB Translate Model	0.5494	0.3570	0.2200	0.1375	0.1344	0.3694	0.2402	0.0852	0.4140
	Mean (μ)	0.5355	0.3500	0.2153	0.1349	0.1315	0.3553	0.2311	0.0877	0.4136
MiniGPT4 [33]	Google Translate API	0.4769	0.2929	0.1789	0.1136	0.1881	0.3491	0.1249	0.1324	0.4132
	Yandex Translate API	0.4831	0.2917	0.1735	0.1078	0.1791	0.3343	0.1028	0.1129	0.4069
	NLLB Translate Model	0.4569	0.2747	0.1639	0.1030	0.1854	0.3416	0.0917	0.1176	0.4056
	Mean (μ)	0.4723	0.2865	0.1721	0.1081	0.1842 ¹	0.3417	0.1065	0.1210 ²	0.4086
LlaVA [34]	Google Translate API	0.2920	0.1557	0.0861	0.0513	0.1473	0.2565	0.0407	0.1069	0.3761
	Yandex Translate API	0.2904	0.1549	0.0820	0.0469	0.1435	0.2334	0.0301	0.0930	0.3706
	NLLB Translate Model	0.2745	0.1416	0.0741	0.0415	0.1414	0.2388	0.0287	0.0943	0.3688
	Mean (μ)	0.2856	0.1507	0.0807	0.0466	0.1441	0.2429	0.0332	0.0981	0.3718
BakLlaVA [35]	Google Translate API	0.4616	0.2611	0.1517	0.0915	0.1395	0.3039	0.1574	0.0978	0.4039
	Yandex Translate API	0.4385	0.2469	0.1394	0.0828	0.1295	0.2646	0.1201	0.0816	0.3947
	NLLB Translate Model	0.4411	0.2434	0.1353	0.0778	0.1352	0.2877	0.1327	0.0843	0.3949
	Mean (μ)	0.4471	0.2505	0.1421	0.0840	0.1347	0.2854	0.1367	0.0879	0.3978
GIT [36]	Google Translate API	0.5506	0.3600	0.2344	0.1577	0.1358	0.3687	0.2556	0.0943	0.4174
	Yandex Translate API	0.5306	0.3516	0.2224	0.1450	0.1281	0.3345	0.1937	0.0767	0.4089
	NLLB Translate Model	0.5723	0.3793	0.2504	0.1709	0.1385	0.3775	0.2555	0.0858	0.4171
	Mean (μ)	0.5512	0.3636	0.2357	0.1578	0.1341	0.3602	0.2349	0.0856	0.4144

Table 4
Model performances on the merged caption set created by combining the captions of all language translators.

Translator	Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	SPICE	WMD
All captions (Google Translate API + Yandex Translate API + NLLB Translate model)	CLIPCap [26]	0.3812	0.2548	0.1538	0.0976	0.0823	0.2603	0.0790	0.0263	0.3556
	BLIP [27]	0.6812 ²	0.5120 ³	0.3796 ³	0.2840 ³	0.1694	0.4489 ³	0.2832 ³	0.0580	0.4542 ³
	BLIP2 [28]	0.6262	0.4712	0.3399	0.2485	0.1582	0.4242	0.2662	0.0532	0.4445
	FUSECAP [29]	0.5729	0.3862	0.2437	0.1577	0.1933 ²	0.3957	0.1350	0.0880 ¹	0.4330
	OFA [30]	0.7097 ¹	0.5389 ¹	0.3940 ¹	0.2875 ¹	0.1797 ³	0.4627 ¹	0.2938 ¹	0.0626	0.4605 ²
	PromptCap [31]	0.6688 ³	0.5124 ²	0.3820 ²	0.2868 ²	0.1726	0.4572 ²	0.2850 ²	0.0593	0.4612 ¹
	Kosmos2 [32]	0.6160	0.4487	0.2988	0.2002	0.1512	0.4087	0.2367	0.0529	0.4348
	MiniGPT4 [33]	0.5536	0.3678	0.2363	0.1554	0.2031 ¹	0.3849	0.1052	0.0847 ²	0.4223
	LlaVA [34]	0.3461	0.2020	0.1159	0.0701	0.1601	0.2768	0.0319	0.0682 ³	0.3839
	BakLlaVA [35]	0.5124	0.3164	0.1938	0.1218	0.1531	0.3322	0.1400	0.0559	0.4160
	GIT [36]	0.6428	0.4710	0.3299	0.2335	0.1548	0.4179	0.2419	0.0516	0.4362

for each image. So, the proposed dataset contains both quite simple and extremely detailed image captions, and therefore the images have much longer descriptions (much more and detailed information) than the traditional databases. This fact may contribute to the models producing more accurate and semantically rich descriptions.

The ground-truth (reference) captions for all images in the Tiny TR-CAP dataset were written by one person. At first glance, this may be seen as a weakness of the proposed dataset regarding the verification of the correctness of the reference captions. However, the fact that the reference captions for all the images were written by a single



Fig. 4. Captions from all models for a set of images.

person yielded some particular advantages. Firstly, it made it easier to verify how different and varied all the captions were from each other and guaranteed the diversity of the captions. The caption texts written by different people for the same image would be more likely to be repetitive. Although this would increase the number of captions, it would significantly affect the diversity of the captions. In addition, selecting appropriate captions from a large pool and checking their correctness would require a significant labor effort. These kinds of challenges have the potential to make it almost impossible to build even a tiny benchmark set of captions.

In our study, we employed translated captions from Turkish to English to measure the performance of different deep learning models. If these translated captions from Tiny TR-CAP are to be used in the training of a model, these translated captions may add noise and inaccuracies to the training process. Although currently available translation systems are quite successful in language translation, in some

cases automatic language translation can produce inaccurate texts. In such cases, external NLP models and methods can also be utilized to eliminate inaccurate translations as noise. By the way, the caption translation process may not preserve well the semantic and cultural nuances necessary for accurate multilingual training. However, we hope that recently proposed deep learning-based language translation models (such as NLLB) will be able to overcome such shortcomings to a large extent in the future. The tools we used in this study may not be able to preserve some semantic and cultural nuances well. However, external experts who may use the Tiny TR-CAP in their research study can quickly review the existing captions and update them according to the cultural and semantic nuances in their own languages. As is known, this is a common challenge for automated captioning systems built for native languages.

Finally, besides general-purpose image captioning, the Tiny TR-CAP dataset can also be used for some domain-specific image captioning

Table 5
Some image-caption datasets in english in the current literature.

Caption dataset	Number of total images	Number of captions per image
IAPR TC-12 [61]	20,000	1–5
PASCAL1K [62]	1000	5
UIUC Pascal [63]	1000	5
SBU dataset [64]	1,000,000	1
Flickr8K [21]	8092	5
Flickr30K [65]	31,783	5
MS COCO [20]	328,000	5
YFCC100M [66]	100,000,000	≥1
Google Refexp [67]	26,711	≥1
InstaPIC [68]	721,176	1
FlickrStyle10K [69]	10,000	1–5
Google's Conceptual Captions [70]	≈3,300,000	1
Nocaps [71]	15,100	11
TextCaps [72]	28,408	5
Conceptual 12M [73]	12,000,000	1

tasks. The model trainings can be performed using the Tiny TR-CAP dataset, and the computed/updated model weights can be transferred to domain-specific image captioning models on a transfer learning basis. However, it is debatable how this would affect the performance of the domain-specific captioning models.

6. Conclusions

In this paper, we introduce a novel image captioning benchmark dataset for general-purpose image captioning tasks, called Tiny TR-CAP. It consists of 1076 original images and 5380 handwritten captions. We also tested the performance of 11 different state-of-the-art deep learning-based caption generator models, including CLIPCap, BLIP, BLIP2, FUSECAP, OFA, PromptCap, Kosmos2, MiniGPT4, LLaVA, Bak-LLaVA, and GIT, on Tiny TR-CAP. In the experimental studies, quite promising success rates were observed, and the best success rates were achieved with the OFA model, with scores of 0.7097 BLEU-1, 0.5389 BLEU-2, 0.3940 BLEU-3, 0.2875 BLEU-4, 0.1797 METEOR, 0.4627 ROUGE-L, 0.2938 CIDEr, 0.0626 SPICE, and 0.4605 WMD. To contribute to the academic research on automatic image captioning, the image and caption sets of Tiny TR-CAP will also be publicly available on GitHub (https://github.com/abbasmemis/tiny_TR-CAP).

CRedit authorship contribution statement

Abbas Memiş: Writing – original draft, Writing – review & editing, Supervision, Resources, Methodology, Investigation, Data curation, Conceptualization. **Serdar Yıldız:** Writing – original draft, Writing – review & editing, Software, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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